

# DETECTION DEPTH OF SPECTRAL DEFECTS IN SHIFTED ZETA STRINGS

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ABSTRACT. We study how a finite-rank spectral defect is detected by a positive Stieltjes–Weyl system. After a resolvent compactification of an unbounded Stieltjes measure at a scale  $s_0 > 0$ , the compactified moments are an invertible binomial transform of the Taylor coefficients of the Weyl function at the negative resolvent point  $-s_0$ . Finite-rank defects perturb the associated Hankel determinants through exact Christoffel–Darboux kernel formulas, and the first loss of Stieltjes positivity is located exactly at the first anomalous coefficient of the associated Stieltjes continued fraction. For a fixed exterior defect over a regular compactly supported background the detection order satisfies  $n_\varepsilon/\log(1/\varepsilon) \rightarrow (2g)^{-1}$ , where  $g$  is the Green function of the complement of the admissible support. We apply this framework to the positive shifted zeta-string reference family associated with Suzuki’s shifted screw functions: a hypothetical zero quartet beyond the shifted boundary enters as a conjugate rank-two defect with effective displacement  $\eta = \delta - \omega$ , the same parameter that controls exponential growth in the time-domain criterion, and the signed absolutely continuous correction it produces is dominated by the reference density and cannot affect the leading detection depth. We derive the resulting inverse-spectral sensitivity law and compare it with the time-domain detection scale  $\eta t_* = 2 \log \tau + \log \log \tau + \log(S_\omega/\eta) + o(1)$ . The results quantify the very different conditioning of two exact analytic representations of the same hypothetical defect; they provide neither a new zero-free region nor any progress toward the Riemann hypothesis.

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## 1. INTRODUCTION

**1.1. The question.** A positive measure  $\sigma$  on  $[0, \infty)$  with  $\int (1 + \lambda)^{-1} d\sigma(\lambda) < \infty$  corresponds, through its Stieltjes–Weyl function

$$q(z) = \int_0^\infty \frac{d\sigma(\lambda)}{\lambda - z}, \quad z \in \mathbb{C} \setminus [0, \infty),$$

to a positive inverse-spectral object: a Kreĭn string, equivalently a positive Stieltjes continued fraction [1, 4, 2, 3]. If the spectral data are corrupted by a small *defect* located outside the admissible support — a point mass on the negative axis, or a conjugate pair of non-real point masses — then positivity of the inverse problem must eventually fail. The question this paper quantifies is:

*How much finite-resolution spectral information is required before a small off-support defect becomes detectable as a failure of positivity?*

Three classical finite-resolution coordinates are available: truncated moment sequences, Taylor jets of the Weyl function at a fixed point of the resolvent set, and initial segments of the Stieltjes continued fraction (equivalently, initial pieces of the string). A first contribution of this paper is that, after a natural one-parameter *resolvent compactification* that makes all moments of an unbounded Stieltjes measure finite, these three coordinates are exactly equivalent, degree by degree (Theorem 2.2). The central mechanism is then:

spectral defect  $\longrightarrow$  finite-rank Hankel perturbation  $\longrightarrow$  Christoffel threshold  $\longrightarrow$  first CF failure.

Finite-rank perturbations of Hankel matrices are controlled exactly by Christoffel–Darboux kernels (Theorems 3.3 and 3.4); the location of the first failed positivity condition translates exactly, through the classical determinant representation of the Stieltjes continued fraction, into the index of the first anomalous continued-fraction coefficient (Theorem 3.1); and for a fixed exterior defect over a regular compactly supported background, exterior polynomial growth turns the threshold into the asymptotic law

$$(1) \quad \frac{n_\varepsilon}{\log(1/\varepsilon)} \longrightarrow \frac{1}{2g(c)} \quad (\varepsilon \downarrow 0),$$

where  $g$  is the Green function of the complement of the admissible support with pole at infinity (Theorem 4.1). We call  $n_\varepsilon$  the *detection depth* of the defect: it is simultaneously a moment order, a Taylor-jet order, and (up to the exact parity rule of Theorem 3.1) the position of the first anomalous bead of the associated string.

**1.2. The arithmetic application.** Suzuki's shifted screw functions [15] attach to each shift  $\omega > 0$  a criterion function  $\Psi_\omega$  whose eventual nonnegativity is equivalent to a zero-free half-plane  $\operatorname{Re} \rho \leq \frac{1}{2} + \omega$  for the Riemann xi function, and, in the companion papers [16, 17], a Weyl function

$$(2) \quad q_\omega(z) = \frac{1}{w} \frac{\xi'}{\xi} \left( \frac{1}{2} + \omega + w \right), \quad w = \sqrt{-z}, \operatorname{Re} w > 0,$$

which, when the corresponding half-plane is zero-free — in particular under the Riemann hypothesis — is a Stieltjes function whose spectral measure is absolutely continuous with an explicit Poisson-broadened zero density. This *positive shifted-zeta reference system* is a structured arithmetic background for the abstract theory: a hypothetical zero  $\rho = \frac{1}{2} + \delta \pm i\tau$  with  $\delta > \omega$  contributes to (2) a conjugate pair of principal-sheet poles at

$$z_\pm = -(\eta \pm i\tau)^2, \quad \eta := \delta - \omega > 0,$$

each of Stieltjes weight 2 (Proposition 5.1) — a rank-two defect in the sense of the abstract theory, whose distance from admissibility is controlled by exactly the *horizontal excess*  $\eta = \delta - \omega$  that also drives the exponential instability  $e^{\eta t}$  of the time-domain criterion. The signed absolutely continuous correction that accompanies the poles is dominated pointwise by the reference density and cannot influence the leading detection depth (Lemma 5.2). The resulting sensitivity law is Theorem 5.4; the time-domain counterpart, with its constant term identified, is

$$(3) \quad \eta t_* = 2 \log \tau + \log \log \tau + \log \frac{S_\omega}{\eta} + o(1), \quad S_\omega = \frac{\xi'}{\xi} \left( \frac{1}{2} + \omega \right),$$

(Theorems 6.1 and 6.2).

**1.3. What is, and is not, claimed.** Three caveats govern the entire paper. First, the zeta application is a *sensitivity statement about the positive reference system perturbed by one hypothetical quartet*; if the Riemann hypothesis fails, the actual response need not decompose as a positive background plus a single finite-rank defect, and we never assume that it does (see the discussion before Theorem 5.4). Second, the compactification is a natural *one-parameter* family indexed by the scale  $s_0 > 0$ ; the detection order is cutoff-independent for fixed  $s_0$  but genuinely  $s_0$ -dependent, and no claim of unique canonicity is made (Section 5.6). Third, the high-height law  $n \asymp \tau^2 \log \tau / (\eta \sqrt{s_0})$  is a moving-point asymptotic: it follows from the fixed-point theorem only formally, and we present it at the level actually established — an explicit geometric computation of the Green distance plus numerical confirmation in the regimes accessible to computation (Section 5.5). The results are statements about the *conditioning of two exact representations* of the same hypothetical object. They provide no new zero-free region and no evidence that the inverse-spectral representation brings the Riemann hypothesis closer.

**1.4. Relation to existing work.** Every individual algebraic ingredient below is classical: the Stieltjes moment problem and continued fraction [1, 2, 5], the Hankel determinant representation of the continued-fraction coefficients [2, 3], finite-rank (Uvarov) modifications of orthogonality measures [6, 7], Christoffel functions and Christoffel–Darboux kernels [8, 9], regularity and  $n$ -th root asymptotics [10], and transformations between Stieltjes and Hausdorff moment problems [11]. The principle that Christoffel-function growth detects points outside the support is also actively used in applied anomaly detection [12, 13, 14]. What we believe is new is the synthesis: the exact identification of the first failed positivity condition with the first anomalous continued-fraction coefficient, giving a precise inverse-spectral meaning to detection depth; the use of the resolvent compactification as a finite-moment/Taylor-jet depth coordinate for unbounded Weyl data, with its exact transform and length invariance; and the application to

the shifted zeta-string family, where the defect parameter coincides with the horizontal excess  $\eta = \delta - \omega$  of the time-domain criterion. A final literature audit for these specific combinations accompanies the submission.

## 2. STIELTJES PRELIMINARIES AND RESOLVENT COMPACTIFICATION

Throughout,  $\sigma$  denotes a positive Borel measure on  $[0, \infty)$  with

$$(4) \quad \int_0^\infty \frac{d\sigma(\lambda)}{1+\lambda} < \infty,$$

and  $q(z) = \int_0^\infty (\lambda - z)^{-1} d\sigma(\lambda)$  its Stieltjes transform, holomorphic on  $\mathbb{C} \setminus [0, \infty)$ . We work with this normalization (no additive affine term); the statements below extend to the general Stieltjes normalization by trivial modifications.

**Definition 2.1** (Resolvent compactification at scale  $s_0$ ). Fix  $s_0 > 0$  and set

$$x = \frac{\lambda}{s_0 + \lambda} \in [0, 1), \quad \zeta = \frac{z}{s_0 + z}, \quad z = \frac{s_0 \zeta}{1 - \zeta}.$$

Let  $\mu_{s_0}$  be the pushforward under  $\lambda \mapsto x$  of the measure  $d\sigma(\lambda)/(s_0 + \lambda)$ , and let

$$Q_{s_0}(\zeta) = \int_0^1 \frac{d\mu_{s_0}(x)}{x - \zeta}.$$

**Theorem 2.2** (Resolvent compactification and Taylor-jet/moment equivalence). *Let  $\sigma$  satisfy (4) and let  $s_0 > 0$ . Then:*

- (1)  $\mu_{s_0}$  is a finite positive measure on  $[0, 1]$ ; in particular all compactified moments  $\hat{\mu}_k = \int_0^1 x^k d\mu_{s_0}(x)$  are finite.
  - (2) The transforms are related by
- $$(5) \quad q(z) = (1 - \zeta) Q_{s_0}(\zeta), \quad \text{equivalently} \quad Q_{s_0}(\zeta) = \frac{s_0 + z}{s_0} q(z).$$
- (3) Writing  $q(z) = \sum_{j \geq 0} c_j (z + s_0)^j$  near  $z = -s_0$ , one has  $c_j = \int_0^\infty (s_0 + \lambda)^{-j-1} d\sigma(\lambda)$  and
- $$(6) \quad \hat{\mu}_k = \sum_{j=0}^k \binom{k}{j} (-s_0)^j c_j.$$

*In particular, for every  $N$ , the moments  $(\hat{\mu}_k)_{k \leq N}$  and the Taylor coefficients  $(c_j)_{j \leq N}$  determine each other through the invertible triangular transform (6).*

*Proof.* (1) is (4) after the change of variables, since  $d\mu_{s_0}$  has total mass  $\int (s_0 + \lambda)^{-1} d\sigma \leq \max(1, s_0^{-1}) \int (1 + \lambda)^{-1} d\sigma < \infty$ .

(2) A direct computation gives

$$x - \zeta = \frac{\lambda}{s_0 + \lambda} - \frac{z}{s_0 + z} = \frac{s_0(\lambda - z)}{(s_0 + \lambda)(s_0 + z)},$$

hence

$$Q_{s_0}(\zeta) = \int_0^\infty \frac{(s_0 + \lambda)(s_0 + z)}{s_0(\lambda - z)} \cdot \frac{d\sigma(\lambda)}{s_0 + \lambda} = \frac{s_0 + z}{s_0} q(z),$$

and  $1 - \zeta = s_0/(s_0 + z)$  gives the first form.

(3) For  $z$  near  $-s_0$ ,

$$\frac{1}{\lambda - z} = \frac{1}{(\lambda + s_0) - (z + s_0)} = \sum_{j \geq 0} \frac{(z + s_0)^j}{(\lambda + s_0)^{j+1}},$$

with uniform convergence on compact subsets of  $|z + s_0| < s_0 \leq \text{dist}(-s_0, [0, \infty))$ ; integrating against  $\sigma$  identifies  $c_j$ . Next,

$$x^k = \left(1 - \frac{s_0}{s_0 + \lambda}\right)^k = \sum_{j=0}^k \binom{k}{j} (-s_0)^j \frac{1}{(s_0 + \lambda)^j},$$

so

$$\hat{\mu}_k = \int_0^\infty x^k \frac{d\sigma(\lambda)}{s_0 + \lambda} = \sum_{j=0}^k \binom{k}{j} (-s_0)^j \int_0^\infty \frac{d\sigma(\lambda)}{(s_0 + \lambda)^{j+1}} = \sum_{j=0}^k \binom{k}{j} (-s_0)^j c_j.$$

The matrix of the transform is lower triangular with diagonal entries  $(-s_0)^k \neq 0$ , hence invertible.  $\square$

*Remark 2.3.* Equation (6) says that the compactification is nothing other than *re-expansion of the Weyl function about an interior point of the resolvent set*: compactified moment order = Taylor-jet order at  $-s_0$ . This is what makes the moment coordinate intrinsic for fixed  $s_0$  — it requires no truncation of the underlying spectral data — and it is also of practical value: high-precision compactified moments of a measure with intractable density are obtained from Cauchy-integral Taylor coefficients of  $q$  on a circle about  $-s_0$ .

**Corollary 2.4** (Equal total string length). *Suppose in addition  $q(0^-) < \infty$ . In the Kreĭn correspondence between Stieltjes functions and strings, normalized so that the total string length equals the value of the Weyl function at  $z = 0$  [4], the strings associated with  $q$  and  $Q_{s_0}$  have the same total length:  $Q_{s_0}(0) = q(0)$  by (5).*

*Remark 2.5* (Non-locality of the bead map). The equality of lengths is global: the compactification does *not* act locally on the mass distribution of the string. For a single bead  $[m, \ell]$  one computes  $[m + 1/\ell, \ell]$ , and for two or more beads the interior masses and gaps mix non-locally (the last mass always maps to  $m_N + 1/\ell_N$ ). We therefore refer to depth in the compactified string as *resolvent-compactified depth at scale  $s_0$* , a bulk coordinate of a canonically associated equal-length string, not a re-parametrization of the original one.

**Proposition 2.6** (Bounded-transform interpretation; finite vector measures). *Let  $A \geq 0$  be self-adjoint on a Hilbert space  $\mathcal{H}$ , let  $\phi \in \mathcal{H}$ , and let  $\sigma = \sigma_\phi$  be the (finite) spectral measure of  $A$  in  $\phi$ . Then  $B := A(s_0 + A)^{-1}$  is a positive contraction,  $0 \leq B < 1$ , and with  $\psi := (s_0 + A)^{-1/2}\phi$ ,*

$$\langle \psi, B^k \psi \rangle = \hat{\mu}_k \quad (k \geq 0).$$

*Proof.* Functional calculus:  $\langle \psi, B^k \psi \rangle = \int \left(\frac{\lambda}{s_0 + \lambda}\right)^k \frac{d\sigma_\phi(\lambda)}{s_0 + \lambda} = \hat{\mu}_k$ .  $\square$

*Remark 2.7* (Infinite total mass). The measures arising in Section 5 have infinite total mass, so they are not spectral measures of ordinary Hilbert-space vectors, and Proposition 2.6 is stated for finite vector measures only. The main development is measure-theoretic and does not use Proposition 2.6. (A rigged-scale formulation is possible, since (4) says exactly that  $(s_0 + A)^{-1/2}\phi$  makes sense for  $\phi$  in the natural negative space; we do not pursue this here.)

### 3. FINITE-RANK SPECTRAL DEFECTS AND THE FIRST CONTINUED-FRACTION FAILURE

**3.1. Hankel determinants and the Stieltjes continued fraction.** For a real sequence  $(\mu_j)_{j \geq 0}$  put

$$\Delta_n = \det(\mu_{i+j})_{i,j=0}^{n-1}, \quad \Delta'_n = \det(\mu_{i+j+1})_{i,j=0}^{n-1}, \quad \Delta_0 = \Delta'_0 = 1.$$

We use throughout the string form of the Stieltjes continued fraction,

$$(7) \quad q(z) = \frac{1}{-zm_1 + \frac{1}{\ell_1 + \frac{1}{-zm_2 + \frac{1}{\ell_2 + \dots}}}},$$

associated with  $q(z) = \int (t - z)^{-1} d\mu(t) = -\sum_{k \geq 0} \mu_k z^{-k-1}$  as a formal expansion at  $z = \infty$ ; in the Kreĭn string picture the  $m_k$  are point masses and the  $\ell_k$  the gaps between them, with the first mass at the measuring end [4, 3]. The classical determinant formulas, in this convention, are

$$(8) \quad m_k = \frac{(\Delta'_{k-1})^2}{\Delta_{k-1} \Delta_k}, \quad \ell_k = \frac{\Delta_k^2}{\Delta'_{k-1} \Delta'_k} \quad (k \geq 1),$$

see [1, 2, 3]; we have re-verified (8) in this exact indexing symbolically for one and two atoms and numerically to  $10^{-112}$  for generic six-atom measures. Positivity of all  $m_k, \ell_k$  up to level  $N$  is equivalent to  $\Delta_n > 0$  and  $\Delta'_n > 0$  up to the corresponding orders — the classical solvability criterion for the Stieltjes moment problem [1, 5].

**Theorem 3.1** (Exact index of the first continued-fraction failure). *Let  $(\mu_j)$  be a real sequence whose Hankel determinants  $\Delta_n, \Delta'_n$  are nonzero up to the orders below. Define*

$$n_0 = \min\{n \geq 1 : \Delta_{n-1} \Delta_n < 0\}, \quad n_1 = \min\{n \geq 1 : \Delta'_{n-1} \Delta'_n < 0\}$$

(with  $\min \emptyset = \infty$ ). *If  $\min(n_0, n_1) < \infty$ , then all coefficients of (7) preceding the following one are positive, and the first nonpositive coefficient is*

$$\begin{cases} m_{n_0}, & \text{raw index } 2n_0 - 2, & \text{if } n_0 \leq n_1, \\ \ell_{n_1}, & \text{raw index } 2n_1 - 1, & \text{if } n_1 < n_0. \end{cases}$$

(Coefficients are indexed  $m_1, \ell_1, m_2, \ell_2, \dots$  with raw indices  $0, 1, 2, 3, \dots$ ; a tie  $n_0 = n_1$  appears first as a mass since  $2n - 2 < 2n - 1$ .)

*Proof.* By (8) the numerators are squares, so  $\text{sgn } m_k = \text{sgn}(\Delta_{k-1} \Delta_k)$  and  $\text{sgn } \ell_k = \text{sgn}(\Delta'_{k-1} \Delta'_k)$ . All factors before the first sign change in each family are positive; the first negative product in either family occurs exactly at  $n_0$ , resp.  $n_1$ , and the interleaving  $m_1, \ell_1, m_2, \ell_2, \dots$  places  $m_{n_0}$  at raw index  $2n_0 - 2$  and  $\ell_{n_1}$  at  $2n_1 - 1$ .  $\square$

**Remark 3.2.** Theorem 3.1 is the bridge between moment-space detection and inverse-string depth: *the first anomalous continued-fraction coefficient is exactly the first order at which the truncated Stieltjes moment problem detects the defect*, and the parity of the anomaly (mass versus length) records which Hankel family failed first. Both statements are convention-checked facts about (8); the content of the following two theorems is *where* that failure occurs.

### 3.2. Rank-one defects: a forbidden real atom.

**Theorem 3.3** (Exact rank-one Christoffel threshold). *Let  $\mu_0$  be a finite positive measure with compact support  $K \subset [0, \infty)$ , with Hankel matrices  $H_n = (\mu_{i+j})_{i,j=0}^{n-1}$  and  $H'_n = (\mu_{i+j+1})_{i,j=0}^{n-1}$  nonsingular, and let*

$$\mu_\varepsilon = \mu_0 + \varepsilon \delta_a, \quad a < 0, \quad \varepsilon > 0.$$

*Then all ordinary Hankel determinants of  $\mu_\varepsilon$  remain positive, while*

$$(9) \quad \frac{\det H'_n(\varepsilon)}{\det H'_n(0)} = 1 + \varepsilon a K'_{n-1}(a, a),$$

where  $K'_{n-1}(a, a) = v(a)^T (H'_n)^{-1} v(a)$ ,  $v(a) = (1, a, \dots, a^{n-1})^T$ , is the (diagonal) Christoffel–Darboux kernel of the positive measure  $t d\mu_0(t)$ . Consequently the first Stieltjes failure is a shifted-Hankel failure, at the exact threshold

$$(10) \quad n_{\text{det}} = \min\{n : \varepsilon |a| K'_{n-1}(a, a) > 1\},$$

and by Theorem 3.1 it appears as the length coefficient  $\ell_{n_{\text{det}}}$ , at raw continued-fraction index  $2n_{\text{det}} - 1$ .

*Proof.*  $\mu_\varepsilon$  is a positive measure on  $\mathbb{R}$ , so its ordinary Hankel matrices are positive semidefinite Gram matrices; with  $H_n(0) > 0$  and  $\varepsilon > 0$  they are positive definite. For the shifted family, the moment increments are  $\delta\mu_k = \varepsilon a^k$ , so  $H'_n(\varepsilon) = H'_n(0) + \varepsilon a v(a)v(a)^T$  is a rank-one update, and the matrix determinant lemma gives  $\det H'_n(\varepsilon) = \det H'_n(0) (1 + \varepsilon a v(a)^T (H'_n)^{-1} v(a))$ . The identification of  $v^T (H')^{-1} v$  with the CD kernel of  $t d\mu_0$  is the classical Christoffel-function identity: if  $(p'_j)$  are the orthonormal polynomials of  $t d\mu_0$ , then  $(H'_n)^{-1} = P^T P$  where  $P$  maps coefficient vectors to  $(p'_j)$ -coordinates, so  $v^T (H'_n)^{-1} v = \sum_{j \leq n-1} p'_j(a)^2 = K'_{n-1}(a, a)$  [8, 9]. Since  $a < 0$  and  $K'_{n-1}(a, a) > 0$  is nondecreasing in  $n$ , the ratio (9) decreases through 0 exactly at (10); the ordinary family never fails, so Theorem 3.1 applies with  $n_1 = n_{\text{det}} < n_0 = \infty$ .  $\square$

### 3.3. Rank-two defects: a conjugate non-real pair.

**Theorem 3.4** (Exact rank-two determinant identity). *Let  $\mu_0$  be as above,  $c \in \mathbb{C} \setminus \mathbb{R}$ , and consider the real moment perturbation*

$$\mu_k(\varepsilon) = \mu_k + \frac{\varepsilon}{2} c^k + \frac{\varepsilon}{2} \bar{c}^k \quad (k \geq 0),$$

*i.e. formal conjugate atoms  $\frac{\varepsilon}{2} \delta_c + \frac{\varepsilon}{2} \delta_{\bar{c}}$ . Define the bilinear kernel  $K_{n-1}^{\text{bil}}(z, w) = v(z)^T H_n^{-1} v(w) = \sum_{j \leq n-1} p_j(z) p_j(w)$  and the Hermitian value  $K_{n-1}(c, \bar{c}) = \sum_{j \leq n-1} |p_j(c)|^2$ , where  $(p_j)$  are the orthonormal polynomials of  $\mu_0$ . Then*

$$(11) \quad \frac{\det H_n(\varepsilon)}{\det H_n(0)} = 1 + \varepsilon \operatorname{Re} K_{n-1}^{\text{bil}}(c, c) + \frac{\varepsilon^2}{4} \left( |K_{n-1}^{\text{bil}}(c, c)|^2 - K_{n-1}(c, \bar{c})^2 \right),$$

*and the coefficient of  $\varepsilon^2/4$  is nonpositive, by Cauchy–Schwarz  $|\sum p_j(c)^2| \leq \sum |p_j(c)|^2$ . The analogous formula holds for the shifted family  $H'_n$ , with kernels of  $t d\mu_0$  and defect columns weighted by  $c, \bar{c}$ .*

*Proof.*  $H_n(\varepsilon) = H_n + UCU^{*T}$  in the bilinear sense, with  $U = [v(c) \ v(\bar{c})]$  and  $C = \operatorname{diag}(\varepsilon/2, \varepsilon/2)$ ; since the perturbation is symmetric (not Hermitian) in the Hankel structure, apply the two-column matrix determinant lemma  $\det(H + UCV^T) = \det H \cdot \det(I_2 + CV^T H^{-1} U)$  with  $V = [v(\bar{c}) \ v(c)]$ , so that the moment increments  $\frac{\varepsilon}{2}(c^{i+j} + \bar{c}^{i+j})$  are reproduced. Writing  $V^T H^{-1} U = \begin{pmatrix} K^{\text{bil}}(\bar{c}, c) & K^{\text{bil}}(\bar{c}, \bar{c}) \\ K^{\text{bil}}(c, c) & K^{\text{bil}}(c, \bar{c}) \end{pmatrix}$  and using that the background moments are real, so that  $K^{\text{bil}}(\bar{c}, \bar{c}) = \overline{K^{\text{bil}}(c, c)}$  and  $K^{\text{bil}}(c, \bar{c}) = K^{\text{bil}}(\bar{c}, c) = K_{n-1}(c, \bar{c}) \in (0, \infty)$ , the  $2 \times 2$  determinant expands to (11). Cauchy–Schwarz in  $\ell^2$  gives the sign of the quadratic term.  $\square$

**Remark 3.5** (No factor-of-two shortcut). The negative channel in (11) grows on the exponential scale like  $\varepsilon^2 e^{4ng(c)}$ , but the defect weight is squared: solving either threshold gives the same leading order  $n \sim \log(1/\varepsilon)/(2g(c))$  at fixed defect location. Observed depth differences between defect types in our laboratory experiments came from different Green distances of the defect locations, not from the exponent  $4g$ .

*Remark 3.6* (Finitely many defects). For  $r$  defects at  $c_1, \dots, c_r$  with weight matrix  $C$ , the same lemma gives  $\det H_n(\varepsilon)/\det H_n = \det(I_r + C\mathcal{K}_n)$  with  $\mathcal{K}_n$  the  $r \times r$  matrix of CD-kernel evaluations at the defect points: an exact finite-dimensional interaction law. On the exponential scale the first failure is governed by the defect(s) of smallest Green distance; we leave the multi-defect asymptotics to future work.

#### 4. CHRISTOFFEL DETECTION DEPTH

Let  $K \subset \mathbb{R}$  be compact, non-polar, with Green function  $g_K$  of  $\overline{\mathbb{C}} \setminus K$  with pole at infinity, and let  $\mu_0$  be a positive measure with  $\text{supp } \mu_0 = K$  that is *regular* in the Stahl–Totik sense [10], i.e. its orthonormal polynomials obey the  $n$ -th root asymptotics  $|p_n(z)|^{1/n} \rightarrow e^{g_K(z)}$  locally uniformly on  $\mathbb{C} \setminus \text{Co}(K)$ , and hence

$$(12) \quad \frac{1}{2n} \log K_n(c, \bar{c}) \longrightarrow g_K(c) \quad (n \rightarrow \infty)$$

for fixed  $c \notin K$ , where  $K_n(c, \bar{c}) = \sum_{j \leq n} |p_j(c)|^2$ . (For real  $a \notin \text{Co}(K)$  the same holds for  $K_n(a, a)$ ; we also apply (12) to the measure  $t d\mu_0$ , which is regular if  $\mu_0$  is and  $0 \notin K$ .)

**Theorem 4.1** (Fixed-defect detection depth). *Let  $\mu_0$  be regular with compact non-polar support  $K \subset (0, \infty)$ .*

(1) (*Rank one.*) *For a defect  $\varepsilon \delta_a$  with fixed  $a < 0$ , the first failure order  $n_\varepsilon$  of (10) satisfies*

$$\frac{n_\varepsilon}{\log(1/\varepsilon)} \longrightarrow \frac{1}{2g_K(a)} \quad (\varepsilon \downarrow 0).$$

(2) (*Rank two.*) *For a conjugate defect pair at fixed  $c \in \mathbb{C} \setminus \mathbb{R}$  as in Theorem 3.4, assume the nondegeneracy condition that the negative channel obeys the root asymptotic  $\frac{1}{4n} \log(K_{n-1}(c, \bar{c})^2 - |K_{n-1}^{\text{bil}}(c, c)|^2) \rightarrow g_K(c)$ . Then the first failure order of the relevant Hankel family satisfies*

$$\frac{n_\varepsilon}{\log(1/\varepsilon)} \longrightarrow \frac{1}{2g_K(c)} \quad (\varepsilon \downarrow 0).$$

*Proof.* (1) Fix  $\varepsilon_0 > 0$ . By (12) applied to  $t d\mu_0$  (also regular, with the same exterior Green function), and since the fixed factor  $|a| > 0$  is absorbed by any exponent tolerance, there is  $n_*$  such that

$$e^{2n(g-\varepsilon_0)} \leq |a| K'_{n-1}(a, a) \leq e^{2n(g+\varepsilon_0)} \quad (n \geq n_*).$$

The threshold (10) is crossed at the first  $n$  with  $\varepsilon |a| K'_{n-1}(a, a) > 1$ ; since  $K'_n$  is nondecreasing in  $n$ , the two-sided bound yields, for all small  $\varepsilon$ ,

$$\frac{\log(1/\varepsilon)}{2(g+\varepsilon_0)} - O(1) \leq n_\varepsilon \leq \frac{\log(1/\varepsilon)}{2(g-\varepsilon_0)} + O(1),$$

and  $\varepsilon_0 \downarrow 0$  gives the claim.

(2) Write (11) as  $1 + \varepsilon L_n - \frac{\varepsilon^2}{4} N_n$  with  $L_n = \text{Re } K_{n-1}^{\text{bil}}(c, c)$ , so  $|L_n| \leq K_{n-1}(c, \bar{c})$ , and  $N_n = K_{n-1}(c, \bar{c})^2 - |K_{n-1}^{\text{bil}}(c, c)|^2 \geq 0$ . Write  $L := \log(1/\varepsilon)$ . *No failure below the window:* for  $n \leq (1-\kappa)L/(2g)$  and  $\varepsilon_0 < g\kappa/(1-\kappa)$ , both  $\varepsilon |L_n| \leq \varepsilon e^{2n(g+\varepsilon_0)} \rightarrow 0$  and  $\varepsilon^2 N_n \leq \varepsilon^2 e^{4n(g+\varepsilon_0)} \rightarrow 0$ , so the ratio stays positive. *Failure above the window:* for  $n \geq (1+\kappa)L/(2g)$  the nondegeneracy hypothesis gives, for  $\varepsilon_0$  small,

$$\frac{\varepsilon^2}{4} N_n \geq \frac{1}{4} e^{-2L+4n(g-\varepsilon_0)} \geq \frac{1}{4} e^{2\kappa' L}, \quad \varepsilon |L_n| \leq e^{-L+2n(g+\varepsilon_0)} \leq e^{(\kappa'+o(1))L}$$

with  $\kappa' > 0$ ; hence  $\varepsilon^2 N_n/(\varepsilon |L_n|) \geq \frac{1}{4} e^{(\kappa'-o(1))L} \rightarrow \infty$ , and the determinant ratio  $1 + \varepsilon L_n - \frac{\varepsilon^2}{4} N_n$  is negative for all small  $\varepsilon$ , whatever the sign of  $L_n$ . Since a monotone-in- $n$  failure witness is not

needed — we only require that no  $n$  below the window fails and some  $n$  above it fails — the first failure order lies in  $[(1 - \kappa), (1 + \kappa)] \cdot L/(2g)$ , and  $\kappa \downarrow 0$  completes the proof.  $\square$

*Remark 4.2* (What regularity does and does not give). Stahl–Totik regularity supplies  $n$ -th root asymptotics and hence the ratio limit of Theorem 4.1. It does *not* supply the multiplicative constants that would be needed for a statement of the form  $n_\varepsilon = \log(C/\varepsilon)/(2g) + O(1)$ ; such refinements require strong (Szegő class) asymptotics and additional hypotheses. Our numerical experiments confirm this caution: the effective constant varies substantially with the background and with the compactification scale (Section 7).

**Corollary 4.3** (Continued-fraction defect depth). *In the setting of Theorem 4.1, the first anomalous continued-fraction coefficient of the defected system occurs at raw index  $2n_\varepsilon + O(1)$  with the mass/length parity of Theorem 3.1: Green distance from the admissible support is, asymptotically, the inverse-spectral resolution depth.*

## 5. THE SHIFTED ZETA-STRING REFERENCE FAMILY

**5.1. The positive reference system.** Let  $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$  and, for  $\omega > 0$ , define  $q_\omega$  by (2). The following facts are established in [15, 16, 17]: if the half-plane  $\operatorname{Re} s > \frac{1}{2} + \omega$  is free of zeros of  $\xi$  — in particular, for every  $\omega > 0$ , under the Riemann hypothesis — then  $q_\omega$  is a Stieltjes function; its spectral measure  $\sigma_\omega$  is absolutely continuous on  $(0, \infty)$  with density

$$(13) \quad \rho_\omega(\lambda) = \frac{1}{\pi\sqrt{\lambda}} \operatorname{Re} \frac{\xi'}{\xi} \left( \frac{1}{2} + \omega - i\sqrt{\lambda} \right) = \frac{1}{\pi\sqrt{\lambda}} \sum_{\gamma} \frac{\omega}{\omega^2 + (\sqrt{\lambda} - \gamma)^2},$$

the sum running over the ordinates of the (paired) zeros: the Poisson-broadened zero measure. The density is continuous and strictly positive on  $(0, \infty)$ , with  $\rho_\omega(\lambda) \sim S_\omega/(\pi\sqrt{\lambda})$  as  $\lambda \downarrow 0$ , where  $S_\omega = (\xi'/\xi)(\frac{1}{2} + \omega) > 0$ , and  $\rho_\omega(\lambda) \asymp \log \lambda/\sqrt{\lambda}$  as  $\lambda \rightarrow \infty$ ; in particular  $\int (1 + \lambda)^{-1} d\sigma_\omega < \infty$ , so Theorem 2.2 applies. We call  $(\sigma_\omega, q_\omega)$  the *positive shifted-zeta reference system*. After compactification at scale  $s_0$ , the reference measure  $\mu_{\omega, s_0}$  is a finite measure on  $[0, 1]$  whose density is continuous and positive on  $(0, 1)$ ; in particular  $\mu_{\omega, s_0}$  is regular in the Stahl–Totik sense [10].

## 5.2. One hypothetical quartet.

**Proposition 5.1** (Location and weight of the defect). *Insert into (2) a hypothetical simple zero  $\rho = \frac{1}{2} + \delta + i\tau$  with  $\delta > \omega > 0$ ,  $\tau > 0$ , together with the conjugate and functional-equation partners  $\bar{\rho}$ ,  $1 - \rho$ ,  $1 - \bar{\rho}$ . Set  $\eta = \delta - \omega > 0$ . On the principal branch  $\operatorname{Re} w > 0$ , the added quartet contributes to  $q_\omega$  exactly two poles, located at*

$$z_\pm = -(\eta \pm i\tau)^2 = \tau^2 - \eta^2 \mp 2i\eta\tau,$$

*each with Stieltjes-form weight 2:  $q_\omega(z) \sim 2/(z_\pm - z)$  as  $z \rightarrow z_\pm$ . Under resolvent compactification at scale  $s_0$  the defect points and weights become*

$$c_\pm = \frac{z_\pm}{s_0 + z_\pm}, \quad \varepsilon_{\text{eff}, \pm} = \frac{2}{s_0 + z_\pm}, \quad |\varepsilon_{\text{eff}}| \sim \frac{2}{\tau^2} \quad (\tau \rightarrow \infty).$$

*The partners  $1 - \rho, 1 - \bar{\rho}$  produce no principal-sheet poles; together with the cut contribution of the quartet they produce a signed absolutely continuous correction treated in Lemma 5.2.*

*Proof.*  $\xi'/\xi(\frac{1}{2} + \omega + w)$  has a simple pole of residue 1 at each zero; the zero  $\rho$  gives  $w_0 = \rho - \frac{1}{2} - \omega = \eta + i\tau$  with  $\operatorname{Re} w_0 > 0$  (principal sheet), and  $\bar{\rho}$  gives  $\bar{w}_0$ ; the partners give  $\operatorname{Re} w < 0$ .

Near  $w_0$ , with  $z = -w^2$ ,  $z - z_+ = -2w_0(w - w_0) + O((w - w_0)^2)$ , so

$$q_\omega(z) = \frac{1}{w} \cdot \frac{1}{w - w_0} (1 + o(1)) = \frac{1}{w_0} \cdot \frac{-2w_0}{z - z_+} (1 + o(1)) = \frac{2}{z_+ - z} (1 + o(1)).$$

The compactified data follow from Theorem 2.2: a pole  $w/(\lambda_0 - z)$  transforms to a pole at  $x_0 = \lambda_0/(s_0 + \lambda_0)$  with weight  $w/(s_0 + \lambda_0)$ , applied at  $\lambda_0 = z_\pm$ .  $\square$

**5.3. The signed cut correction.** Beyond its two principal poles, the quartet alters the boundary values of  $q_\omega$  on the cut  $[0, \infty)$ : writing the quartet's exact contribution to  $(\xi'/\xi)(\frac{1}{2} + \omega + w)$  as  $d(w) = \sum_{\gamma_v \in \{\tau - i\delta, \tau + i\delta\}} \left( \frac{1}{\omega + w - i\gamma_v} + \frac{1}{\omega + w + i\gamma_v} \right)$ , the function  $q_{\text{def}}(z) = d(\sqrt{-z})/\sqrt{-z}$  decomposes as the two poles of Proposition 5.1 plus an absolutely continuous part on  $[0, \infty)$  with signed density

$$(14) \quad \rho_{\text{def}}(\lambda) = \frac{1}{\pi\sqrt{\lambda}} \operatorname{Re} d(-i\sqrt{\lambda}).$$

When the defect is scaled by a small parameter  $\varepsilon$  (the sensitivity model), the correction is  $\varepsilon\rho_{\text{def}}$ .

**Lemma 5.2** (The cut correction cannot affect the leading depth). *Fix  $\omega > 0$ ,  $s_0 > 0$  and the quartet parameters  $(\tau, \delta)$ ,  $\delta > \omega$ . Then*

$$M_* := \sup_{\lambda > 0} \frac{|\rho_{\text{def}}(\lambda)|}{\rho_\omega(\lambda)} < \infty.$$

Consequently, for  $0 < \varepsilon < 1/(2M_*)$ , the perturbed background  $d\sigma_\omega + \varepsilon\rho_{\text{def}} d\lambda$  satisfies, in quadratic form on every polynomial subspace and for every  $n$ ,

$$(1 - \varepsilon M_*) H_n^{(0)} \leq H_n^{(0)} + \varepsilon R_n \leq (1 + \varepsilon M_*) H_n^{(0)},$$

where  $H_n^{(0)}$  and  $R_n$  are the compactified Hankel matrices of  $\sigma_\omega$  and of the cut correction. Hence the measure “background + cut correction” remains a positive measure with the same support, regularity class, and Green function, its Christoffel kernels are distorted by factors in  $[(1 + \varepsilon M_*)^{-1}, (1 - \varepsilon M_*)^{-1}]$ , and the first failure order of the full defected system equals that of the principal-pole (rank-two) model up to an  $O(1)$  shift as  $\varepsilon \downarrow 0$ :

$$n_\varepsilon^{\text{full}} = n_\varepsilon^{\text{pole}} + O(1) \quad (\varepsilon \downarrow 0).$$

In particular the ratio limit of Theorem 4.1 is unchanged.

*Proof. Boundedness of the ratio.* Both densities are continuous and positive (resp. continuous) on  $(0, \infty)$ , so it suffices to compare the ends. As  $\lambda \downarrow 0$ :  $\operatorname{Re} d(-i\sqrt{\lambda})$  tends to the finite constant  $d(0)$ , while  $\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \omega - i\sqrt{\lambda}) \rightarrow S_\omega > 0$ ; both densities behave like  $\text{const}/\sqrt{\lambda}$  and the ratio tends to  $|d(0)|/S_\omega$ . As  $\lambda \rightarrow \infty$ : with  $u = \sqrt{\lambda}$ ,  $\operatorname{Re} d(-iu) = O(u^{-2})$  (two Lorentzian pairs centred at  $\pm\tau$ ), while  $\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \omega - iu) \asymp \log u \rightarrow \infty$  by (13) and the Riemann–von Mangoldt density; the ratio tends to 0. Hence  $M_* < \infty$ .

*Quadratic-form domination.* For any polynomial  $P$  (in the compactified variable; the compactification map is a fixed smooth bijection, so pointwise domination of densities is preserved),

$$\left| \int P^2 \varepsilon \rho_{\text{def}} \right| \leq \varepsilon M_* \int P^2 \rho_\omega = \varepsilon M_* \langle P, H^{(0)} P \rangle.$$

*Threshold stability.* Let  $\tilde{H}_n = H_n^{(0)} + \varepsilon R_n$ ; by the two-sided bound,  $\tilde{H}_n$  is positive definite for  $\varepsilon < 1/M_*$  and its inverse satisfies  $(1 + \varepsilon M_*)^{-1} (H_n^{(0)})^{-1} \leq \tilde{H}_n^{-1} \leq (1 - \varepsilon M_*)^{-1} (H_n^{(0)})^{-1}$ , hence the CD kernels obey  $\tilde{K}_n(c, \bar{c}) \in [(1 + \varepsilon M_*)^{-1}, (1 - \varepsilon M_*)^{-1}] \cdot K_n(c, \bar{c})$ , and the bilinear values are controlled by the same two-sided operator bound via polarization, uniformly in  $n$ , once

$\varepsilon \leq 1/(2M_*)$ . The exact rank-two threshold of Theorem 3.4, computed against  $\tilde{H}_n$  instead of  $H_n^{(0)}$ , is therefore displaced by at most a fixed multiplicative factor  $1 + O(\varepsilon)$  inside the logarithm, i.e. the first failure order shifts by  $O(\varepsilon)/(2g) + O(1) = O(1)$ .  $\square$

*Remark 5.3.* Lemma 5.2 discharges what [18] flagged as the one remaining local proof obligation: the finite-rank model is not an approximation but the exact leading mechanism, the a.c. correction being dominated by the reference density. The same argument applies verbatim to any finite collection of quartets.

**5.4. The sensitivity theorem.** We emphasize the logical status of the following statement: it concerns the positive reference system (2)–(13) perturbed by *one hypothetical quartet scaled by  $\varepsilon$* . It is a perturbative sensitivity theorem about an exactly defined model. It is *not* a statement about the actual zeta function in a world where RH fails: there, nothing would confine the violation to a single quartet, and we make no such claim. (A conditional variant — assuming all remaining zeros admissible, the actual response decomposes as reference plus the quartet’s exact contribution, to which the same analysis applies by Lemma 5.2 — holds with identical proof.)

**Theorem 5.4** (Spectral-defect sensitivity of the shifted-zeta reference system). *Fix  $\omega > 0$ ,  $s_0 > 0$ , and quartet parameters  $\tau > 0$ ,  $\eta = \delta - \omega > 0$ . Let  $\mu_{\omega, s_0}$  be the compactified reference measure,  $g_{\omega, s_0}$  the Green function of the complement of its support, and  $c_{\pm}$  the compactified defect points of Proposition 5.1. Scale the quartet by  $\varepsilon \downarrow 0$  and let  $n_{\varepsilon}$  be the first compactified Hankel order at which the defected system loses Stieltjes positivity. Then, under the nondegeneracy condition of Theorem 4.1(2) at  $c_+$ ,*

$$\frac{n_{\varepsilon}}{\log(1/\varepsilon)} \longrightarrow \frac{1}{2g_{\omega, s_0}(c_+)},$$

*and the first anomalous continued-fraction coefficient of the compactified string occurs at the index given by Theorem 3.1.*

*Proof.* The compactified reference measure is finite with support  $\overline{(0, 1)} = [0, 1]$  and positive continuous density on  $(0, 1)$ , hence regular [10]; Proposition 5.1 realizes the scaled quartet as the conjugate rank-two defect  $\frac{\varepsilon'}{2}\delta_{c_+} + \frac{\varepsilon'}{2}\delta_{c_-}$  with  $\varepsilon' = 2\varepsilon/(s_0 + z_+)$ , plus the cut correction; Lemma 5.2 reduces to the pure finite-rank model; Theorems 3.4 and 4.1(2) give the ratio limit; Theorem 3.1 locates the continued-fraction anomaly.  $\square$

**5.5. High height: a moving-point asymptotic.** For  $\tau \rightarrow \infty$  at fixed  $\eta, s_0, \omega$ , the defect point approaches the compactified endpoint:

$$1 - \operatorname{Re} c_+ \sim \frac{s_0}{\tau^2}, \quad |\operatorname{Im} c_+| \sim \frac{2s_0\eta}{\tau^3},$$

and the equilibrium density of the limiting interval near its upper edge gives the explicit Green asymptotic

$$(15) \quad g(c_+) \sim \frac{2\eta\sqrt{s_0}}{\tau^2} \quad \left( \text{more precisely } 2\eta\sqrt{s_0 + \lambda_{\min}}/\tau^2 \text{ for a support hull } [\lambda_{\min}, \infty) \right),$$

verified against the exact interval Green function to 0.4% at  $\tau = 160$  [19]. Combining (15) with  $|\varepsilon_{\text{eff}}| \sim 2/\tau^2$  *formally* in the fixed-point threshold yields the high-height prediction

$$(16) \quad n_{\text{comp}} \asymp \frac{\tau^2 \log \tau}{2\eta\sqrt{s_0}}.$$

We state (16) as a *moving-point asymptotic supported numerically*: both the evaluation point and the effective weight vary with  $\tau$ , and fixed-point Stahl–Totik regularity does not supply the required uniformity. Establishing uniform Christoffel asymptotics in the joint regime  $n \asymp \tau^2 \log \tau / (\eta \sqrt{s_0})$  — by strong asymptotics for the compactified  $\omega$ -density, endpoint asymptotics plus analytic continuation, Riemann–Hilbert methods, or extremal polynomial bounds — is an open problem we consider tractable but do not resolve here. The numerical section documents both the regime where (16)-type predictions hold (broadening or displacement exceeding the local zero spacing) and the quasi-discrete regime of genuine low-height zeta data, where the interval law overestimates the observed depth.

**5.6. Dependence on the scale  $s_0$ .** The resolvent compactification is a one-parameter family, and the detection order genuinely depends on the scale: in controlled experiments with fixed spectral data and defect,  $n_{s_0}$  is non-monotone in  $s_0$  (e.g. 22, 14, 12, 17, 28 at  $s_0 = 1, 4, 25, 100, 400$  for a fixed moderate-height defect [19]), and no power-law renormalization removes the dependence at moderate scales. At high height with  $\tau^2 \gg s_0$ , (15) gives the leading covariance  $n_{s_0} \propto 1/\sqrt{s_0 + \lambda_{\min}}$ . We therefore use throughout the terminology *resolvent-compactified detection depth at scale  $s_0$* : it is cutoff-independent for fixed  $s_0$  (Theorem 2.2 makes it a Taylor-jet order), but it is not scale-free. The observed minimum of  $n_{s_0}$  near  $s_0 \sim \tau^2$  suggests a detection-optimal scale  $s_0^*(\tau)$ ; we leave this as a future-work observation.

## 6. TIME-DOMAIN DETECTION

We now compare with the shifted screw-function criterion of [15, 16]. For fixed  $\omega > 0$  the positive reference background has the linear margin

$$\Psi_\omega^{\text{bg}}(t) = S_\omega t + O(1), \quad S_\omega = \frac{\xi'}{\xi} \left( \frac{1}{2} + \omega \right) > 0,$$

and one violating quartet contributes the growing oscillation

$$(17) \quad \Psi_{\omega, \rho}^{\text{grow}}(t) = -\frac{2e^{\eta t}}{\tau^2 + \eta^2} \cos(\tau t + \alpha) + O(t),$$

with amplitude coefficient  $2/(\tau^2 + \eta^2)$ ; this normalization has been audited symbolically and numerically against the exact spectral series (the two growing branches are  $e^{(\eta \pm i\tau)t}/(\eta \pm i\tau)^2$ ; agreement of (17) with the exact quartet term to relative  $10^{-8}$  by  $t = 150$ ) [19, 16].

**Theorem 6.1** (Envelope detection time). *Define  $t_{\text{env}}$  by  $\frac{2e^{\eta t_{\text{env}}}}{\tau^2 + \eta^2} = S_\omega t_{\text{env}}$ . Then, for fixed  $\eta, \omega > 0$ , as  $\tau \rightarrow \infty$ ,*

$$\eta t_{\text{env}} = 2 \log \tau + \log \log \tau + \log \frac{S_\omega}{\eta} + o(1).$$

*Proof.* Set  $x = \eta t_{\text{env}}$ ; the defining equation reads  $2e^x/(\tau^2 + \eta^2) = (S_\omega/\eta)x$ , i.e.

$$x = \log(\tau^2 + \eta^2) + \log x + \log \frac{S_\omega}{2\eta}.$$

Since  $\log(\tau^2 + \eta^2) = 2 \log \tau + o(1)$  and a first iteration gives  $x = 2 \log \tau + O(\log \log \tau)$ , we get  $\log x = \log \log \tau + \log 2 + o(1)$ , and substituting back,  $x = 2 \log \tau + \log \log \tau + \log 2 + \log \frac{S_\omega}{2\eta} + o(1) = 2 \log \tau + \log \log \tau + \log \frac{S_\omega}{\eta} + o(1)$ .  $\square$

**Theorem 6.2** (First-negative-time bracket). *Let  $\Psi(t) = S_\omega t + \Psi_{\omega, \rho}(t)$  be the one-quartet perturbative model with the exact quartet term, and  $t_{\text{first}} = \inf\{t > 0 : \Psi(t) < 0\}$ . Assume the transversality condition that at  $t_{\text{env}}$  the envelope crosses the (exact, non-oscillatory) margin*

at the rate  $\eta + o(1)$ , and that the non-oscillatory part varies by  $o(1)$  relative to its value on intervals of length  $O(1/\tau)$ . Then, for  $\tau$  large,

$$0 \leq t_{\text{first}} - t_{\text{env}} \leq \frac{2\pi}{\tau} + o(1/\tau), \quad \text{hence} \quad \eta t_{\text{first}} = 2 \log \tau + \log \log \tau + \log \frac{S_\omega}{\eta} + o(1).$$

*Proof.* Before  $t_{\text{env}}$  the envelope is below the margin, so  $\Psi > 0$ :  $t_{\text{first}} \geq t_{\text{env}}$ . Within  $(t_{\text{env}}, t_{\text{env}} + 2\pi/\tau]$  the cosine attains a trough  $\cos = -1$ ; over that interval the envelope changes by the factor  $e^{2\pi\eta/\tau} = 1 + O(1/\tau)$  and the non-oscillatory part by  $o(1)$  relative terms, so at the trough  $\Psi \leq S_\omega t (1 + o(1)) - \text{envelope}(t) (1 - O(1/\tau)) < 0$  by transversality. Numerically the bracket is comfortable: gaps 0.115, 0.027, 0.011, 0.0035 against bounds  $2\pi/\tau = 0.314, 0.079, 0.020, 0.0049$  at  $\tau = 20, 80, 320, 1280$  [19].  $\square$

**Corollary 6.3** (The two coordinates compared). *For the same hypothetical quartet at height  $\tau$  with horizontal excess  $\eta = \delta - \omega$ : the time-domain criterion detects it at*

$$t_{\text{first}} = \frac{2 \log \tau + \log \log \tau + O(1)}{\eta},$$

by Theorems 6.1–6.2, while the resolvent-compactified inverse-spectral coordinate at scale  $s_0$  requires depth  $n_\varepsilon \sim \log(1/\varepsilon)/(2g_{\omega, s_0}(c_+))$  for the scaled defect (Theorem 5.4), with the high-height moving-point behavior (16) predicting  $n \asymp \tau^2 \log \tau / (2\eta\sqrt{s_0})$ . The time-domain coordinate resolves a high defect on a logarithmic-in-height scale; the compactified moment description is predicted, and numerically observed in the accessible regimes, to require polynomially-in-height larger resolution. This is a statement about the conditioning of two exact representations of the same object, in the stated coordinates and asymptotic regimes.

## 7. NUMERICAL ILLUSTRATIONS

All computations below are reproducible from the laboratory record [19]; multi-precision arithmetic (60–450 digits), exact integer certification where applicable, and control rows (undefected backgrounds remaining Stieltjes-clean) accompany every experiment. Numerics here illustrate regimes and constants; no proof depends on them.

**7.1. Exactness certificates.** The binomial transform (6) was verified on rational toy measures to  $8 \cdot 10^{-131}$ ; the rank-one identity (9) to  $10^{-285}$ ; the rank-two identity (11) to  $10^{-273}$ ; the determinant formulas (8) to  $10^{-112}$ ; and the index law of Theorem 3.1 exactly on twenty synthetic single-defect measures and four graph-Laplacian-type specimens (first anomaly  $2n_1 - 1$  as a length, resp.  $2n_0 - 2$  as a mass, in every case).

**7.2. The a.c. reference at  $\omega > 0$ : interval law at all displacements.** For a scaled reference family (mock ordinates  $\gamma_j/5$ , Poisson broadening  $\omega \in \{0.15, 0.3\}$ ) with one quartet, measured detection orders match the interval Green prediction with no fitted constants across effective displacements  $\eta \in [0.2, 1.2]$  (ratios 0.64–1.01 over eight cases), including displacements far below the ordinate spacing, where the unbroadened (pure-point) background deviates; two parameter pairs with equal  $\eta$  and different  $(\delta, \omega)$  give equal depth, confirming that the defect enters only through the horizontal excess.

**7.3. Unscaled zeta data and the quasi-discrete regime.** With the first 100 genuine ordinates plus the Riemann–von Mangoldt tail,  $s_0 = 200$ ,  $\omega \in \{0.1, 0.2\}$  (control clean to  $n = 28$ ):

| $\omega$ | $\tau$ | $\eta$ | $n_0$ | $n_1$ | $g$ exact | $ \varepsilon_{\text{eff}} $ | measured/naive |
|----------|--------|--------|-------|-------|-----------|------------------------------|----------------|
| 0.1      | 16     | 1.0    | 14    | 14    | 0.18085   | 0.00438                      | 0.93           |
| 0.1      | 20     | 1.0    | 14    | 14    | 0.09379   | 0.00333                      | 0.46           |
| 0.1      | 25     | 1.0    | 22    | 21    | 0.05866   | 0.00242                      | 0.41           |
| 0.1      | 16     | 0.5    | 19    | 18    | 0.09272   | 0.00439                      | 0.61           |
| 0.1      | 20     | 0.5    | 26    | 25    | 0.04705   | 0.00333                      | 0.41           |
| 0.2      | 20     | 1.0    | 16    | 16    | 0.09379   | 0.00333                      | 0.53           |

Detection at genuine zeta scale is finite and computable; the sub-unity ratios reflect the *quasi-discrete regime*: at these heights the local zero spacing  $2\pi/\log(\tau/2\pi) \approx 4.5\text{--}5.4$  exceeds both  $\eta$  and  $\omega$ , and the interval law overestimates the depth. In controlled scaled experiments the ratio approaches 1 once the displacement exceeds roughly 1.5–3 local spacings.

**7.4.  $s_0$ -dependence and the time-domain constant.** The  $s_0$  panel ( $n_{s_0} = 22, 14, 12, 17, 28$  at  $s_0 = 1, 4, 25, 100, 400$ , fixed data) documents the genuine scale dependence of Section 5.6. For the time domain, the solved envelope thresholds converge to the constant of Theorem 6.1: residuals  $\eta t_* - 2\log \tau - \log \log \tau = -3.757 \rightarrow -3.483$  over  $\tau = 10^2\text{--}10^{12}$  against  $\log(S_\omega/\eta) = -3.480$  at  $\omega = 0.1, \eta = 0.15$ .

## 8. DISCUSSION AND OPEN PROBLEMS

**8.1. What has been shown.** An exact chain — finite-rank defect  $\rightarrow$  Hankel perturbation  $\rightarrow$  Christoffel threshold  $\rightarrow$  first continued-fraction failure — with a Green-function resolution law at fixed defect, and an arithmetic realization in which the defect parameter of the shifted zeta family is the same horizontal excess  $\eta = \delta - \omega$  that drives the time-domain criterion. The two exact representations of the same hypothetical violation have parametrically different conditioning: logarithmic in height for the screw function, polynomial in height (in the regimes established) for the compactified inverse-spectral coordinate.

**8.2. What has not been shown.** No new zero-free region; no reformulation that makes the Riemann hypothesis more accessible; no claim that an actual failure of RH would consist of a single quartet over a positive background. The moving-point law (16) awaits uniform Christoffel asymptotics; the constants in the depth law are provably not universal (Remark 4.2, Section 5.6).

### 8.3. Open problems.

*Open problem 8.1* (Endpoint uniformity). As  $\omega \downarrow 0$  the reference measure degenerates from absolutely continuous to pure point, and the observed behavior crosses over from the interval law to a discrete-gap regime governed by the ratio of the defect displacement to the local zero spacing. Prove a uniform (double-scaling) Christoffel asymptotic covering this degeneration.

*Open problem 8.2* (Moving-point uniformity). Establish (16) as a theorem, e.g. through strong asymptotics for the compactified  $\omega$ -density or extremal polynomial bounds uniform near the endpoint.

*Open problem 8.3* (Optimal scale). Characterize the minimizer  $s_0^*(\tau)$  of  $n_{s_0}$  and its asymptotics.

Further directions: multiple-defect interaction laws (Remark 3.6) and their rate-tie phenomena; analogous sensitivity laws for other completed  $L$ -functions, where the same construction applies verbatim to the corresponding shifted families; and the finite (graph zeta) laboratory in which the present mechanism was first isolated, which raises separate questions about non-backtracking spectra and twisted responses and will be treated elsewhere.

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